Capstone Project – Cover Sheet

ASSIGNMENT

Instruction:

- Marks will be allocated based on the quality of presentation, innovation and the depth of your approach.
- Referencing Code: When incorporating code or ideas derived from external sources (such as books, journals, the internet, or discussion forums), proper citation and referencing according to APA conventions within the source code are mandatory. Failure to reference code appropriately will be treated as plagiarism.
- Please ensure that you complete this cover sheet and **affix it to the first page of your project.**
- This project is intended for groups comprising 6 to 7 students.

Student declaration:

I declare that:

- *We understand what is meant by plagiarism*
- *The implication of plagiarism have been explained to us by our lecturer*
- *This project is all our work and we have acknowledged any use of the published or unpublished works of other people.*

Group Leader's Signature: Teo Jun Jie ..... Date: 12 January 2026 .....

GROUP NUMBER: GROUP 7

PROJECT TITLE: Customizable E-learning Platform Tailored to Individual Learning Styles

INTAKE: UCDF2405ICT(SE)

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1.0 Workload Matrix

Area	Teo Jun Jie TP079880	Ng Tian Xin TP081807	Wong Kelvin TP081586	Victor Chia Bing Chen TP080934	Yap Min Xuan TP080104	Lim Xin Ying TP081765
Introduction		50%				50%
Problem & solution		50%				50%
Project Planning					100%	
Feasibility Study	100%					
Systems Analysis			100%			
Design Diagram	16.7%	16.7%	16.7%	16.7%	16.7%	16.7%
Interface Design				100%		
Overall Contribution	**16.7%**	**16.7%**	**16.7%**	**16.7%**	**16.7%**	**16.7%**
Digital Signature	JJ	Vera	Kelvin	Victor	Ann	XinYing

2.0 Introduction

The Evolution of Digital Learning

The landscape of education has undergone a paradigm shift since the 1960s, culminating in the widespread adoption of Massive Open Online Courses (MOOCs) in 2008 (Korableva et al., 2019). This milestone democratized education, enabling educators to deliver digital learning to vast, diverse audiences across the globe. Modern MOOCs typically offer a standardized suite of teaching methods, including pre-recorded videos, static reading materials, and pre-set quizzes (Nguyen, 2022). By aggregating high-quality resources and instructors, these platforms have established themselves as essential tools for self-directed study and review.

The Shift to Hyper-Personalization

However, the post-pandemic era has drastically altered student expectations. Learners have become increasingly reliant on digital platforms, driving a demand for higher service standards. Heavily influenced by the algorithmic personalization found in social media, Generation-Z users now expect similar adaptivity in educational tools (Paik, 2024). They require systems that leverage data analytics to provide interactive pathways tailored to individual preferences and ability levels. This shift highlights a critical gap, while traditional MOOCs excel at content dissemination, they often fail to address individual learning differences, leading to a "one-size-fits-all" experience that can diminish student engagement.

Theoretical Framework: The VARK Model

To bridge this gap, educational theorists advocate for framework like VARK (Visual, Aural, Read/Write, and Kinaesthetic). Developed by Neil Fleming, this model emerged from the observation that students often struggle not because of the content itself, but due to a misalignment between the instruction method and their preferred mode of reception. Empirical studies have demonstrated a statistically significant improvement in academic performance when teaching models are aligned with students' VARK preferences ($p < 0.05$). This evidence suggests that integrating such theoretical frameworks into modern digital platforms is the next logical step to enhance teaching quality and student satisfaction (Melhem & Al-Zoubi, 2025).

2.1 Project Background

The Edume Initiative

In response to the limitations of traditional MOOCs outlined above, this project proposes **"Edume"**, a web-based customizable e-learning platform. As technology and education converge, accessibility alone is no longer sufficient, students aim to absorb the most effective knowledge in the least amount of time. Edume addresses the critical lack of personalization in conventional platforms by operationalizing the VARK model into a functional software system.

Operational Mechanism

The core philosophy of Edume is to maximize learning efficiency, allowing students to achieve "half the work, double the results". The system operates through a structured adaptive mechanism:

1. **Assessment:** Upon entry, students engage with a digital VARK questionnaire.
2. **Analysis:** The system identifies the user's dominant profile:
    - **Visual:** Preference for charts, graphs, and symbolic representations.
    - **Aural:** Preference for audio, discussions, and speaking.
    - **Read/Write:** Preference for text-based input and output.
    - **Kinaesthetic:** Preference for practical exercises, simulations, and real-world practice.
3. **Adaptation:** The platform automatically restructures content delivery. For instance, a Visual learner will see video tutorials prioritized, whereas a Read/Write learner will be presented with comprehensive documentation.

Specialized Feature for IT Education

Recognizing that programming requires practical application, Edume goes beyond passive content. A key innovation of this project is the integration of a **"Simulated Coding Sandbox"** specifically for Kinaesthetic learners. This feature addresses the "practice gap" in online coding courses, offering a risk-free environment for students to apply syntax and receive immediate logic-based feedback.

Our Core Purpose Driven by the need to revolutionize the learning experience, our platform is guided by the following principles:

- **Mission:** To democratize coding education by adapting the curriculum to the student's unique cognitive style.
- **Vision:** To redefine technical education through a data-driven, personalized learning platform.

2.2 Problem Context

The transition to digital education has exposed critical structural flaws in current e-learning models. Despite the accessibility of platforms like MOOCs, three specific challenges hinder effective learning, particularly in technical domain:

2.2.1 The "One-Size-Fits-All" Content Mismatch

The primary deficiency in many existing Learning Management System (LMS) is the reliance on static, standardized content delivery. The research by (Kuraekha & Charoenroop, n.d.) highlights that poor content design and unsuitable teaching methods are prevalent issues. Platforms typically force all students to consume the same textbook-heavy PDFs or lengthy lecture videos. This approach ignores individual cognitive differences. For instance, a visual learner may struggle to conceptualize algorithms through text alone, while a read/write learner may find video lectures inefficient. As noted by (du Plooy et al., 2024), without personalization, this misalignment leads to diminished motivation and suboptimal academic performance.

2.2.2 The "Practise Gap" in Technical Education

Compounding the issues of content format is the significant lack of practical application, a critical component in STEM and coding education (Nguyen, 2022). Traditional MOOCs facilitate passive consumption are watching and reading but frequently fail to provide environments for active experimentation. (Korableva et al., 2019) argue that while these platforms are effective for theory, they do not support the continuous, hands-on practise required to master programming syntax and debugging. For Kinaesthetic learners specifically, the inability to apply knowledge immediately results in a superficial understanding, creating a "practice gap" where students understand the *concept* but cannot write the *code* (Melhem & Al-Zoubi, 2025).

2.2.3 Digital Fatigue and Disengagement

Even when content and practice are available, the static nature of interaction often leads to "Digital Engagement Fatigue". (Maloney et al., 2023) define this as a decrease in learner interest caused by prolonged, monotonous online workloads. Long periods of passive activity, such as consecutive video watching, tax cognitive resources and reduce emotional involvement. When a system lacks dynamic feedback mechanisms, students experience isolation and mental exhaustion, which are key predictors of course dropout.

2.3 Proposed Solution

To directly address the interconnected challenges of personalization, practice, and engagement, the Edume platform implements a three-pillared solution strategy:

2.3.1 VARK-Based Adaptive Content Engine

To resolve the content mismatch, Edume integrates a dynamic profiling mechanism grounded in the VARK model. Upon registration, users complete a structured assessment to identify their dominant learning style (Visual, Aural, Read/Write, or Kinaesthetic). The system's algorithm then reconfigures the curriculum delivery to match this profile (Kuraekha & Charoenroop, n.d.). For example, a Visual learner is prioritized with infographic summaries and video tutorials, whereas a Read/Write learner is presented with structured documentation. This alignment ensures that the *method* of delivery enhances rather than hinders the student's comprehension, a strategy supported by (du Plooy et al., 2024).

2.3.2 Interactive Simulated Coding Sandbox

Bridging the "Practice Gap," Edume introduces a Simulated Coding Sandbox tailored for Kinaesthetic engagement. Unlike passive video players, this feature allows students to write, test, and debug code immediately after learning a concept. The sandbox utilizes a client-side code editor with syntax highlighting and a logic-matching console to provide risk-free, instant validation. This feature ensures that learners can transition from theory to practice within the same environment, reinforcing retention through "learning-by-doing".

2.3.3 Multi-Modal Design with Immediate Feedback Loop

To combat digital fatigue, the platform employs an interactive, multi-modal interface design. By breaking content into segmented, varied formats, switching between watching, reading, and doing, Edume maintains high cognitive arousal and reduces boredom (Akpen et al., 2024). Furthermore, the system incorporates an immediate feedback loop for all quizzes and coding tasks. As noted by (Deng, 2021), timely feedback creates a sense of progression and emotional connection, keeping learners motivated and actively engaged with the material.

2.4 Project Objectives

The project aims to achieve the following specific, measurable objectives:

1. **To develop an automated Learning Style Assessment**
   To design and implement a digital questionnaire based on the VARK methodology that processes user responses in real-time. The system will automatically classify students into Visual, Aural, Read/Write, or Kinaesthetic profiles and store these tags to trigger the adaptive content filtering algorithm.

2. **To implement a "Simulated Coding Sandbox" for Kinaesthetic Engagement:**
   To construct an interactive, web-based coding environment using syntax highlighting libraries for example, CodeMirror that provides immediate, keyword-based feedback. This objective specifically targets Kinaesthetic learners, enabling them to engage in "learning by doing" through guided programming exercises, thereby enhancing practical skill acquisition.

3. **To design a Data Analytics Dashboard for Educational Insights:**
   To create a centralized visualization dashboard for administrators. This system will aggregate user activity data to generate descriptive statistical reports (e.g., student distribution by learning style, content engagement rates), fulfilling the requirement for data-driven insights into how different learning styles correlate with academic progress.

2.5 Project Scope

2.5.1 User Module (Student Interface)

- **Authentication & Profiling:** A secure login system incorporating the mandatory **"VARK Assessment Quiz"** to establish the user's learning profile upon registration.
- **Adaptive Content Player:** A dynamic interface that prioritizes course materials based on the user's profile. Visual learners will be served YouTube-embedded tutorials, while Read/Write learners will access structured text from sources like W3Schools.
- **Interactive Coding Sandbox:** A specialized practice area for programming courses for example, Python Basics. It features a code editor with syntax highlighting and a simulated console that validates user input against pre-set logic patterns to provide instant feedback.

2.5.2 Admin Module (Educator & Administrator)

- **Content Management System (CMS) with Tagging:** An interface allowing administrators to upload heterogeneous resources like videos, text, quizzes, etc. Crucially, admins must assign a **"Learning Style Tag"** to each resource, enabling the system's adaptive filtering logic.
- **Analytics Dashboard:** A visual reporting tool displaying key metrics, such as the percentage of users per VARK category and module completion rates, facilitating data-driven decision-making.

2.5.3 Out of Scope (Exclusions) To ensure project feasibility within the development timeline:

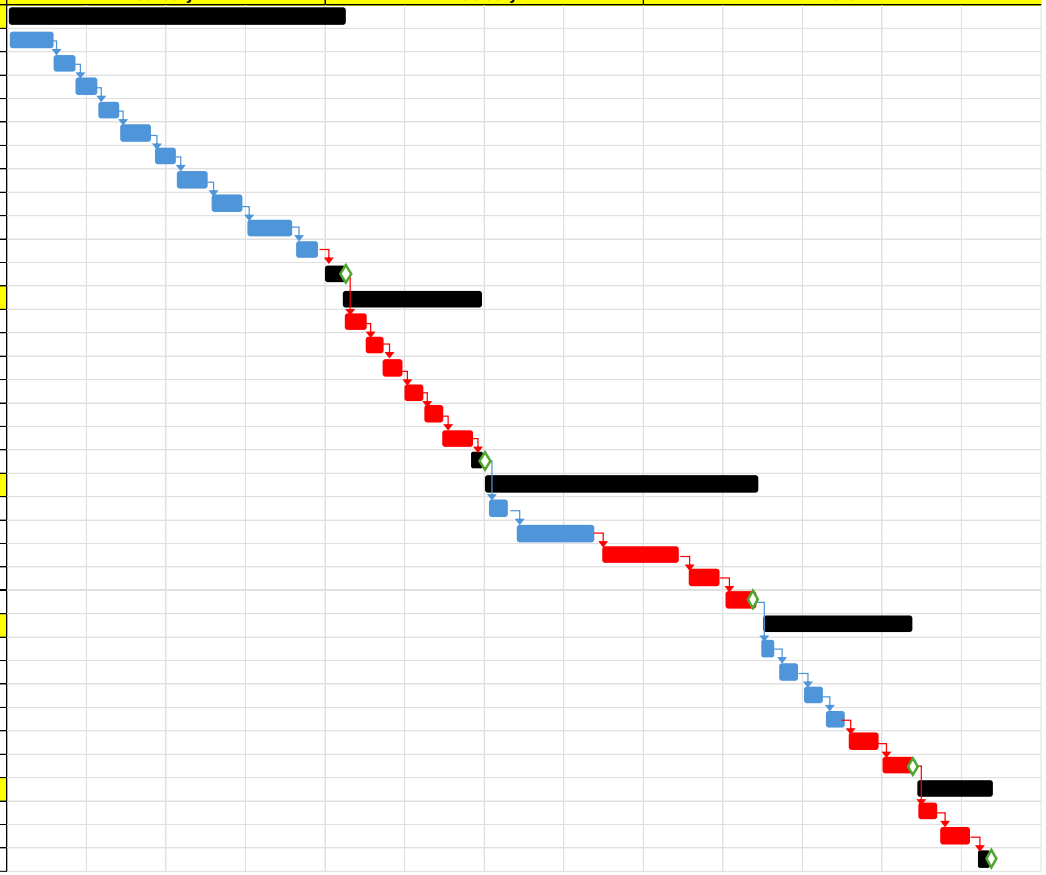
- **Real-time Code Compilation:** The Coding Sandbox uses client-side logic matching for example, pattern checks to simulate execution. It does not include a server-side compiler to avoid security risks and infrastructure complexity.
- **Native Mobile Apps:** The system is a Responsive Web Application accessible via browsers, excluding standalone iOS or Android apps.
- **Live Communication:** Real-time features such as live chat, video conferencing, or peer-to-peer forums are excluded.

3.0 Project Plan

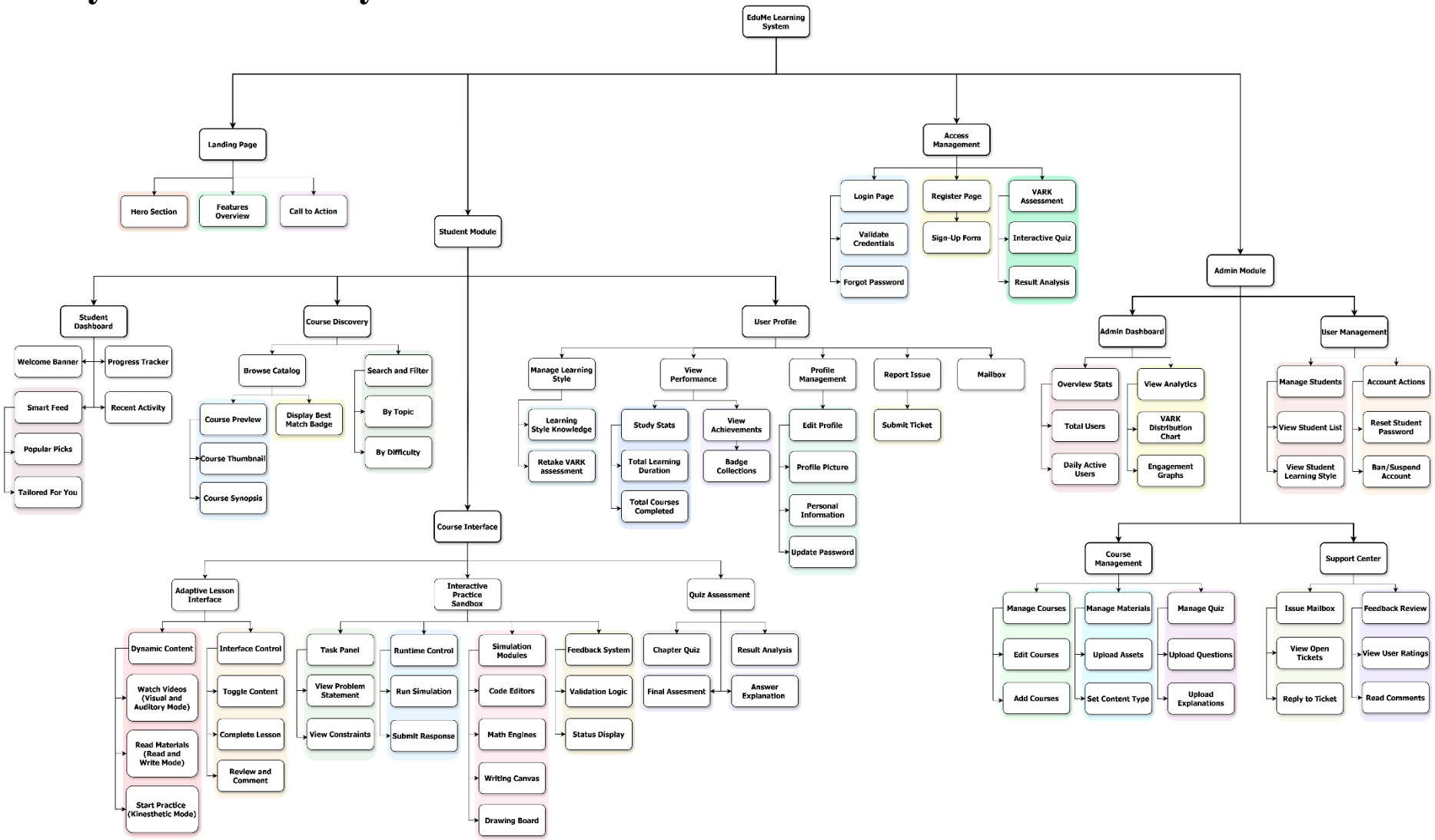
3.1 Project Gantt Chart

Project Gantt Chart

Task Name	Start Date	End Date	Duration (Days)
System Planning and Requirements Analysis			
Background Research	5-Jan	8-Jan	4
Identify Problem	9-Jan	10-Jan	2
Propose Problem Solution	11-Jan	12-Jan	2
Define Project Objectives	13-Jan	14-Jan	2
Define Project Scope	15-Jan	17-Jan	3
Stakeholder Identification	18-Jan	19-Jan	2
Feasibility Study	20-Jan	22-Jan	3
Requirements Gathering	23-Jan	25-Jan	3
Project Planning	26-Jan	29-Jan	4
System Hierarchy Chart	30-Jan	31-Jan	2
Project Proposal	1-Feb	2-Feb	2
System Design			
Context Diagram	3-Feb	4-Feb	2
Data Flow Diagrams	5-Feb	6-Feb	2
Data Dictionary	7-Feb	8-Feb	2
Entity Relationship Diagram	9-Feb	10-Feb	2
Psuedocode & Flowcharts	11-Feb	12-Feb	2
Interface Design	13-Feb	15-Feb	3
Review & Finalization	16-Feb	16-Feb	1
System Implementation			
Hardware & Software Specifications	17-Feb	18-Feb	2
Frontend Developemnt	20-Feb	26-Feb	7
Backend Development	27-Feb	5-Mar	7
API Development	6-Mar	8-Mar	3
Integration & Documentation	9-Mar	11-Mar	3
System Testing			
Test Plan	12-Mar	12-Mar	1
Unit Testing	13-Mar	14-Mar	2
Integration Testing	15-Mar	16-Mar	2
User Acceptance Testing	17-Mar	18-Mar	2
Bug Tracking & Reporting	19-Mar	21-Mar	3
Fixes & Optimize	22-Mar	24-Mar	3
Prototype and Evaluation			
Interactive Prototype	25-Mar	26-Mar	2
Refinement	27-Mar	29-Mar	3
Finalization	30-Mar	30-Mar	1



4.0 System Hierarchy



5.0 Conclusion

Due to everyone being unique, each student may have their own special way of learning because of individual cognitive differences. Edume aims to revolutionize the education scene by changing standard digital learning into a personalized learning environment built on each student's preferred preferences. As traditional teaching methods are static and standardized, Edume solves the issue by using the VARK theoretical framework to analyse student tendencies, in hopes of bringing out a student's full potential through a tailored learning space. In addition, Edume's simulated coding sandbox also provides a safe space for students who prefer a more practical approach in solving real world issues. In a nutshell, Edume hopes to maximize student learning efficiency through their dynamic learning style and responsive content processing algorithms, encouraging students to learn diligently.

6.0 References

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7.0 Appendix

Meeting No:	1		
Date of Meeting:	9th January 2026	Start Time:	4:31pm
Location:	APU Campus, B-08-06	End Time:	4.57pm
Chairperson:	Teo Jun Jie	Minute Taker:	Lim Xin Ying
1. Attendance			
Present			
- Lim Xin Ying - Ng Tian Xin - Teo Jun Jie - Victor Chia Bing Chen - Wong Kelvin - Yap Min Xuan			
Absent		Reason	
- None		- None	
2. Main Points			
Topic/Discussion notes			Discussed By
• Brainstorming and title discussion • Chosen title: **Customizable E-learning Platform Tailored to Individual Learning Styles** • Task delegation			All members

Meeting No:	2		
Date of Meeting:	14th January 2026	Start Time:	4:42pm
Location:	APU Campus, B-08-06	End Time:	5:17pm
Chairperson:	Teo Jun Jie	Minute Taker:	Lim Xin Ying
3. Attendance			
Present			
- Lim Xin Ying - Ng Tian Xin - Teo Jun Jie - Victor Chia Bing Chen - Wong Kelvin - Yap Min Xuan			
Absent		Reason	
- None		- None	
4. Main Points			
Topic/Discussion notes			Discussed By
• Share findings regarding the project • Project name: **Edume** (Education & Me) • Discussion on project problems, solutions and scope • Schedule and track task with Gantt chart			All members

Meeting No:	3		
Date of Meeting:	22nd January 2026	Start Time:	4:24pm
Location:	APU Campus, B-08-06	End Time:	5:01pm
Chairperson:	Teo Jun Jie	Minute Taker:	Lim Xin Ying
5. Attendance			
Present			
- Lim Xin Ying - Ng Tian Xin - Teo Jun Jie - Victor Chia Bing Chen - Wong Kelvin - Yap Min Xuan			
Absent		Reason	
- None		- None	
6. Main Points			
Topic/Discussion notes			Discussed By
• Follow up on progress • Project built using the **VARK model framework** • Discussion regarding system hierarchy chart			All members

Meeting No:	4		
Date of Meeting:	27th January 2026	Start Time:	4:45pm
Location:	APU Campus, B-08-06	End Time:	5:08pm
Chairperson:	Teo Jun Jie	Minute Taker:	Lim Xin Ying
7. Attendance			
Present			
- Lim Xin Ying - Ng Tian Xin - Teo Jun Jie - Victor Chia Bing Chen - Wong Kelvin - Yap Min Xuan			
Absent		Reason	
- None		- None	
8. Main Points			
Topic/Discussion notes			Discussed By
• Finalizing proposal document			All members